string indexing

Forward indexing

In [1]:
```
s1 = 'welcome'
s1
```

Out[1]:
```
'welcome'
```

In [2]:
```
s1[0]
```

Out[2]:
```
'w'
```

In [3]:
```
s1[4]
```

Out[3]:
```
'o'
```

In [4]:
```
s1[10] # string not have that index
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[4], line 1
----> 1 s1[10]

IndexError: string index out of range
```

backward indexing

In [6]:
```
s = 'welcome'
s
```

Out[6]:
```
'welcome'
```

In [7]:
```
s[-3]
```

Out[7]:
```
'o'
```

In [8]:
```
s[-10]  # index not exit
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
Cell In[8], line 1
----> 1 s[-10]  # index not exit

IndexError: string index out of range
```

In [9]:
```
s[-4]
```

Out[9]:
```
'c'
```

step indexing

In [10]:
```
s
```
Out[10]:
```
'welcome'
```

In [11]:
```
s[2:3:1]
```
Out[11]:
```
'l'
```

In [12]:
```
s[:9:-1]
```
Out[12]:
```
''
```

In [13]:
```
s[0:10:2]
```
Out[13]:
```
'wloe'
```

string slicing

In [14]:
```
s = 'welcometeam'
```

In [15]:
```
s[2:10]
```
Out[15]:
```
'lcometea'
```

In [16]:
```
s[:10]
```
Out[16]:
```
'welcometea'
```

In [17]:
```
s[4:]
```
Out[17]:
```
'ometeam'
```

In [18]:
```
s[1:10:3]
```
Out[18]:
```
'eot'
```

In [19]:
```
s[::1]
```
Out[19]:
```
'welcometeam'
```

In [20]:
```
s[::2]
```
Out[20]:
```
'wloeem'
```

In [21]:
```
s[::-1]
```
Out[21]:
```
'maetemoclew'
```

In [22]:
```
s[::-2]
```

Out[22]: 'meeolw'

In [23]:
```
s[:]
```

Out[23]: 'welcometeam'

In [24]:
```
s[::3]
```

Out[24]: 'wcea'

In [25]:
```
s[-9:-1:3]
```

Out[25]: 'lme'

forward slicing

In [26]:
```
s = 'welcometeam'
s
```

Out[26]: 'welcometeam'

In [27]:
```
s[-10:-1:2]
```

Out[27]: 'ecmta'

In [28]:
```
s[3:10:2]
```

Out[28]: 'cmta'

In [29]:
```
s = "PYTHON"
print(s[1:5])
```

```
YTHO
```

backward slicing

In [30]:
```
s = 'welcometeam'
s
```

Out[30]: 'welcometeam'

In [31]:
```
s = "PYTHON"
print(s[::-1])
```

```
NOHTYP
```

In [32]:
```
s = 'welcometeam'
s
```

Out[32]: 'welcometeam'

In [33]:
```
s[-9:-1:3]
```

Out[33]: 'lme'

step slicing

In [34]:
```
s = 'welcometeam'
s
```

Out[34]: 'welcometeam'

In [35]:
```
s[::3]
```

Out[35]: 'wcea'

In [36]:
```
s[::-1]
```

Out[36]: 'maetemoclew'

In [ ]: